

Firm-Dynamics Diagnostics for the E-Cigarette Supply-Side Model

Pre-specification cuts from the Nielsen RMS panel — items 7.1, 7.2, 7.4, 7.5 of the Aug 27 model note (7.3 and 7.6 deferred). Companion to the two 08_05_2026 notes.

August 27, 2026

1. What this note does

The Aug 27 model note (§7) lists six statistics to pull from the Nielsen RMS panel before locking down functional forms for the Experience-Based Equilibrium (EBE) supply model. This note answers four of them — 7.1 (does the cap of five firms bind), 7.2 (entry/exit hazards), 7.4 (a learning-by-doing cost signature), and 7.5 (product-offering-set dynamics and a nicotine-weighting check). Items 7.3 (nicotine vs. FDA event study) and 7.6 (ownership vs. own-quality as the driver of turnover) are deferred and are not addressed here.

Everything is national and monthly (§6.5 of the model note), built from the corrected-nicotine panel — `bt_month_t2_niccorr_from_full.RData` for the brand-month series (7.1, 7.2, 7.4) and a matching UPC-month build for 7.5. The `UNKNOWN` residual bucket (~1.9% of tracked mL, un-attributed to a brand) is dropped from every firm count, entry/exit statistic, and regression. Where a statistic needs a weight, the demand-side notes use mL-sales weights; per Marc's instruction the firm-level nicotine *state* in 7.5 is instead built UPC-unweighted, and the note shows both.

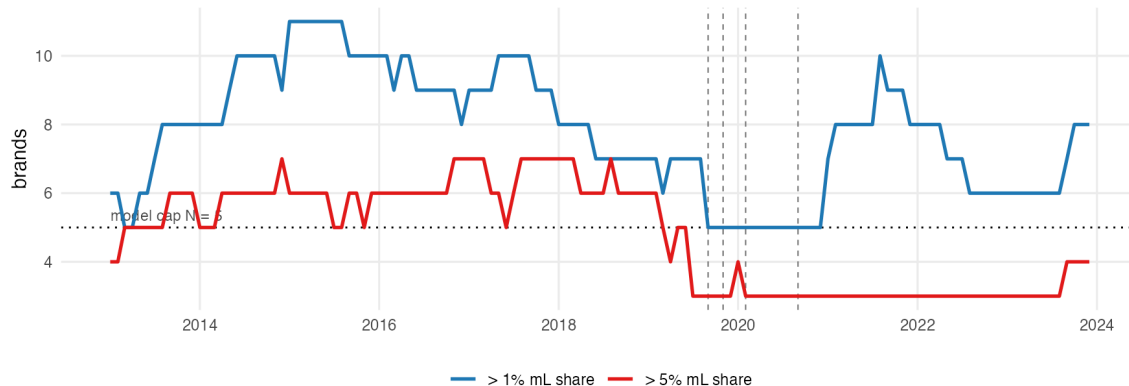
The same data-quality caveats that run through the two 08_05_2026 notes apply here and are flagged in place: the mid-2019 open-refill clearance artifact, and the post-2020 undercoverage of modern disposables (Puff Bar, Elf Bar, Breeze, Lost Mary) that sell mainly through vape shops and online. The second one matters most for 7.2 — a large part of the 2021 "entry" wave is the scanner panel starting to capture brands that were already selling elsewhere, not genuine de novo entry.

Figures and the underlying CSVs are in `output/prelim_analysis/firm_dynamics_ebe/`; the code is `code/analysis/10_ebe_diagnostics.R` and `code/build/5_build_upc_month_niccorr.R`.

7.1 How many brands compete, and does the cap of five bind?

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(a) Number of national brands above a materiality threshold, by month



(b) Aggregate mL share of the fringe (all brands outside the top 5)



Events: 1 EVALI scare (fall 2019) 2 JUUL pulls fruit/mint (Nov 2019)
3 FDA flavored-cartridge enforcement (Feb 2020) 4 PMTA deadline (Sep 2020)

Monthly count of national brands above a materiality threshold, and the aggregate mL share of the fringe outside the top five. Dashed lines mark the timeline events from the 08_05_2026 notes.

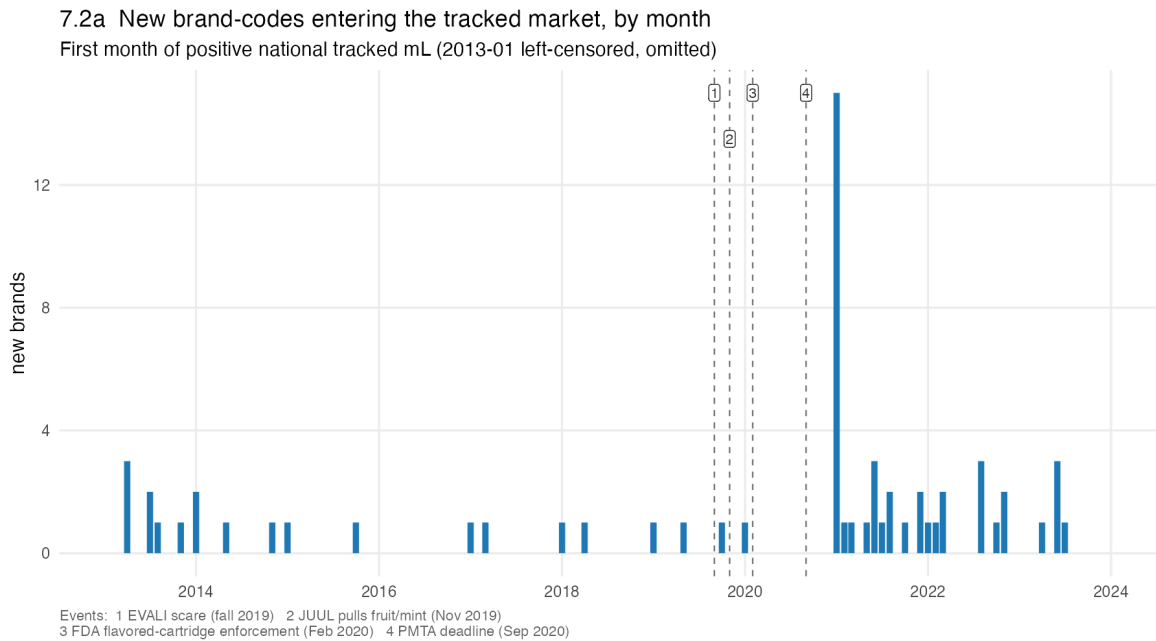
In the median month, 22 brands record positive tracked mL (range 12–39), 8 hold more than 1% of national mL (range 5–11), and 5 hold more than 5% (range 3–7). More than five brands clear the 1% bar in **114 of 132 months** (86%), and more than five clear the 5% bar in **59 of 132 months** (45%).

The 5%-threshold count is strongly era-dependent, and this is the substantive point. Through the fragmented 2014–2018 period, 6–7 brands routinely hold above 5% each and the fringe below the top five carries 15–20% of national mL (peaking near 22% in 2018). The count then falls to 3 across 2020–2023 as the market concentrates into the VUSE/JUUL/NJOY group, and the fringe collapses to about 1% of mL in 2020 before recovering to 6–9% in 2021–2023 as tracked disposables return. The only stretch where exactly five brands clear even the 1% bar is 2019–2020.

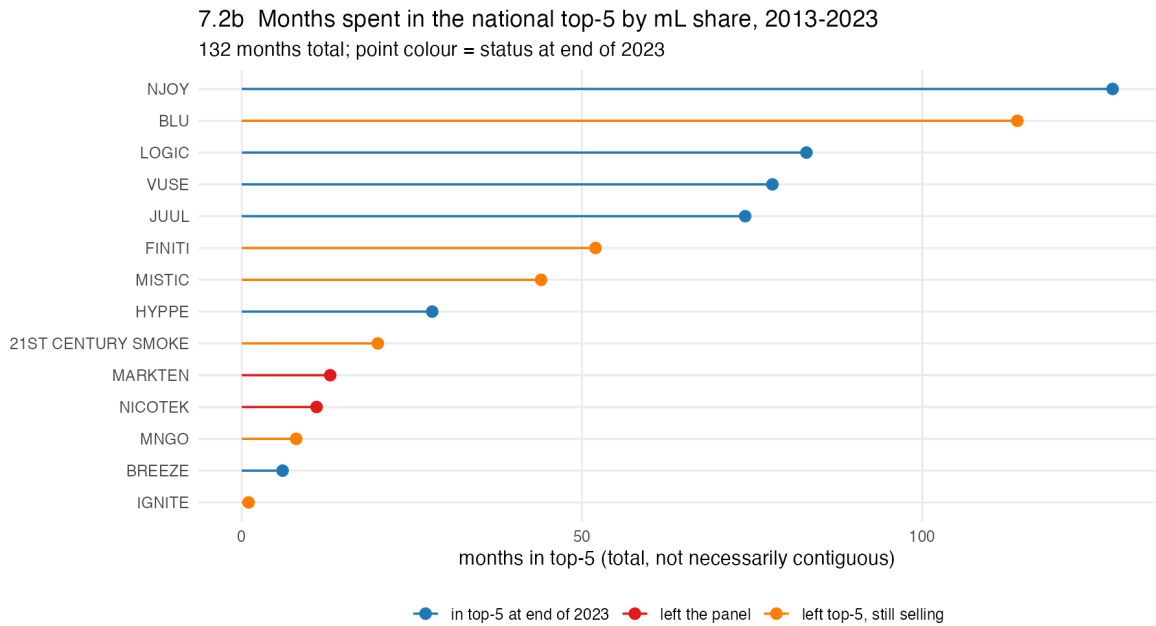
Verdict. Five is not a real cap on the number of economically active firms — it is a reasonable description of the *concentrated* market from 2019 on, and a clear undercount for 2014–2018. Two

implications for the spec. First, if the model is calibrated to the pod era (2019–2023), $\bar{N} = 5$ is defensible and the fringe can plausibly be dropped or folded into the outside option. Second, if the 2013–2018 fragmentation is in the estimation sample, either $\bar{N}$ needs to be larger or the sub-5% brands need to be modeled as a competitive fringe outside the dynamic game — a 15–20% mL residual spread across 4–6 brands is not a rounding error, and its share moves enough year to year (0.4% to 22%) that treating it as a fixed constant would distort the pricing subgame.

7.2 Entry / exit hazards and the one-entrant-per-period assumption



New brand-codes by first month of positive national tracked mL (2013-01 is left-censored and omitted).



Months spent in the national top five by mL share, 2013–2023, by brand and end-of-sample status.

Entry (market-wide). Sixty-three brand-codes first appear after January 2013, spread across 35 distinct months, with up to 15 in a single month. The distribution is very uneven: **27 of the 63 arrive in calendar 2021** alone (the modern-disposable wave — Esco Bars, Fume, and similar), with another 10 in 2022. Before 2021, entry is sparse — 7 new brands in 2013 as scanner coverage ramped, then 0 to 4 per year through 2020 (and 0 in 2016). So "at most one potential entrant per period" is a defensible approximation for the cigalike and pod eras (2014–2020) but is decisively violated in 2021–2022. The caveat cuts in the model's favor: much of the 2021 spike is the scanner panel beginning to capture disposable brands that were already selling through untracked channels (Appendix B of the brand-dynamics note), so it overstates true de novo entry. If the estimation sample stops around 2020, the single-entrant assumption is close to the data; if it runs through 2023, the entry process needs to allow a burst.

Top-five tenure. Fourteen brands reach the national top five by monthly mL share at some point. The median brand spends **36 months** there (mean 47). Six are still in the top five at the end of 2023 (NJOY, LOGIC, VUSE, JUUL, plus the disposables HYPPE and BREEZE); NJOY is in the top five for 128 of the 132 months (out only around its 2016 bankruptcy), and BLU accumulates 114 top-five months before dropping out for good in mid-2022.

Age at exit, and corporate events. Eight brands leave the top five for good. Age (months since first appearance) at the last top-five month is bimodal: quick flame-outs of under-capitalized value brands — 21st Century Smoke (19 months), Nicotek (23), and the one-month 2021 disposable blips MNGO (7) and Ignite (15) — versus slow declines of former leaders — FINITI (51), MarkTen (65), MISTIC (77), BLU (114). Cross-referencing the six exits with meaningful tenure against the dated corporate events in the brand-dynamics note:

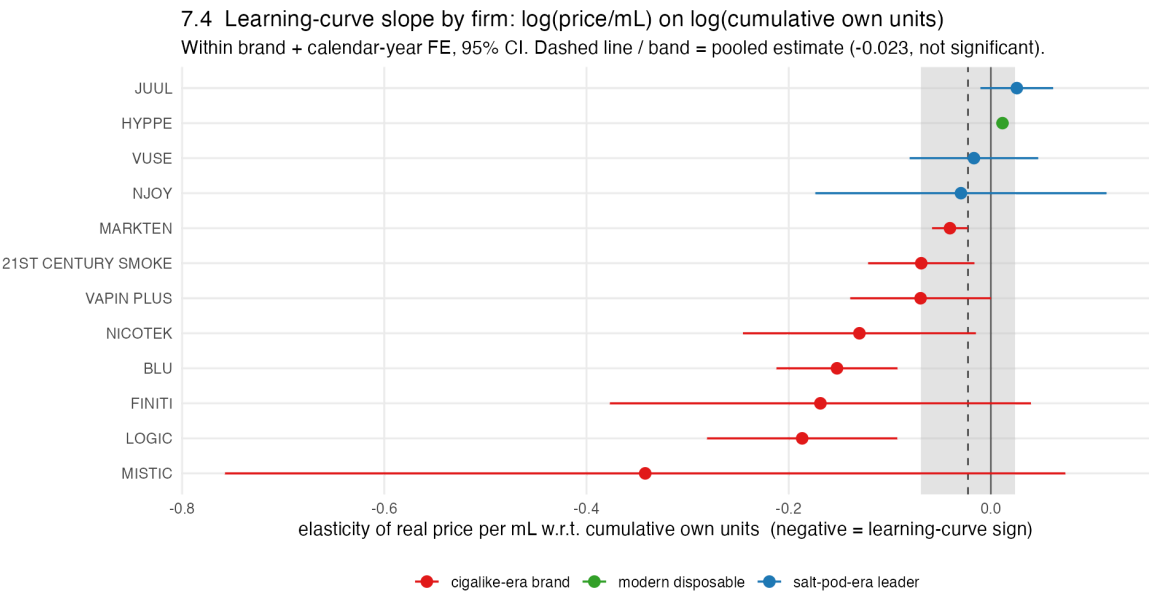
Brand	Last top-5 month	Left panel	Dated corporate event	Read
MarkTen	Jan 2019	Jul 2022	Altria discontinued MarkTen, 2018 (after the Dec 2018 JUUL stake)	coincides — ownership decision, not competitive exit
MISTIC	Jun 2019	trace to 2023	Ballantyne wind-down ~2020; 30 mL bottle liquidation mid-2019	coincides — strategic wind-down (and no competitive pod)

Brand	Last top-5 month	Left panel	Dated corporate event	Read
BLU	Jul 2022	still selling	divested to Imperial, 2015	no coincidence — the divestiture is 7 years earlier; this is a slow bleed
FINITI	Apr 2017	Sep 2023	none nearby	clean competitive exit
21st Century Smoke	Aug 2014	still selling	none nearby	clean competitive exit
Nicotek	Dec 2014	May 2023	none nearby	clean competitive exit

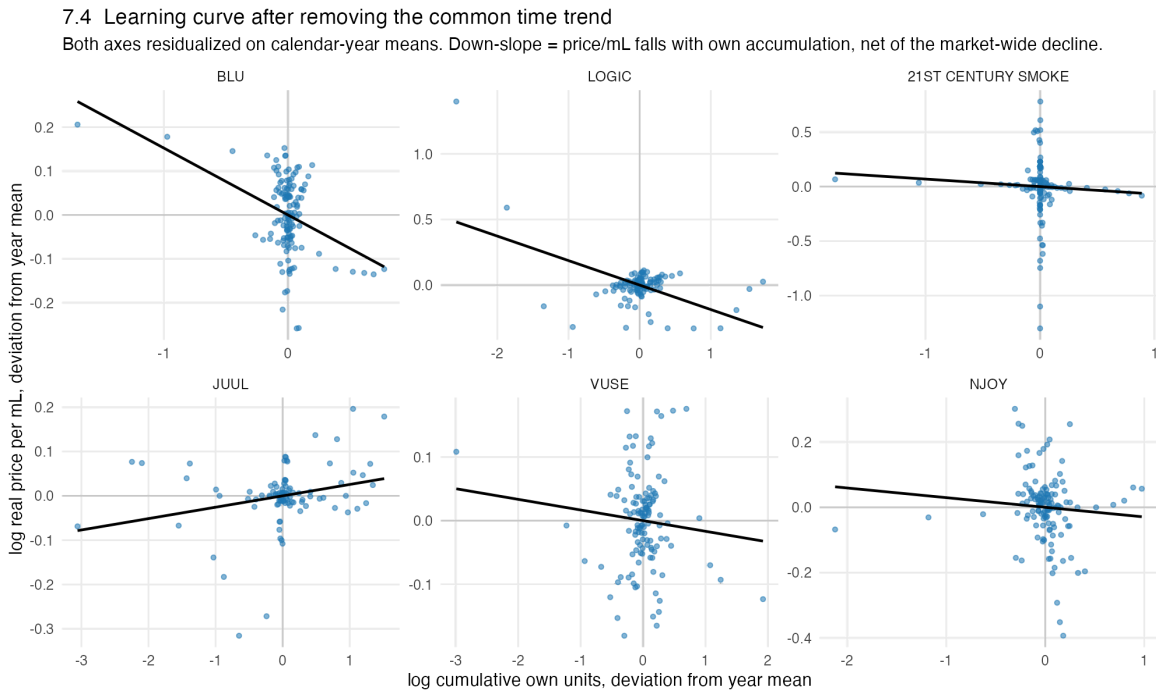
Two of the six line up with a dated corporate event (MarkTen, MISTIC); the other four look like state-dependent competitive decline — gradual for BLU, faster for the small cigalike brands (FINITI, 21st Century, Nicotek). And the three largest *ownership* shocks in the sample — BLU→Imperial (2015), Altria→JUUL (2018 in, 2023 out), Altria→NJOY (2023) — produced **no** top-five exit near the event: JUUL and NJOY are both still in the top five at the end of 2023, and BLU's ranking exit is seven years after its divestiture.

Verdict. The exit side is a workable mix for the model: enough clean competitive exits (FINITI, 21st Century, Nicotek, and the quick disposable flame-outs) to identify a state-dependent exit rule, but a non-trivial minority (MarkTen, MISTIC) where the exit is really a parent-company decision. Those two are candidates to treat as exogenous ownership shocks rather than endogenous exit draws. Note also that RMS brand-codes persist at trace volume long after the brand stops being a going concern (MISTIC "sells" through 2023), so "left the panel" is a soft date — the last-top-five month is the cleaner exit marker. This bears on 7.6 (deferred): the ownership-vs-quality question is live, but on the exit margin specifically the competitive story is the more common one.

7.4 Is there a learning-by-doing cost signature?



Firm-by-firm elasticity of real price per mL with respect to cumulative own units, within brand and calendar-year fixed effects, 95% CI. Dashed line and grey band = pooled estimate.



The same relationship after residualizing both variables on calendar-year means — the variation the firm-by-firm regression uses.

The test regresses log real price per mL for firm i in month t on the log of firm i 's cumulative own units sold through $t - 1$, with brand fixed effects and a time control. Price per mL is revenue over mL, not marginal cost, so this is a reduced-form check for the classic learning-curve sign (own price falling as the firm accumulates output, net of the market-wide trend), not a cost estimate.

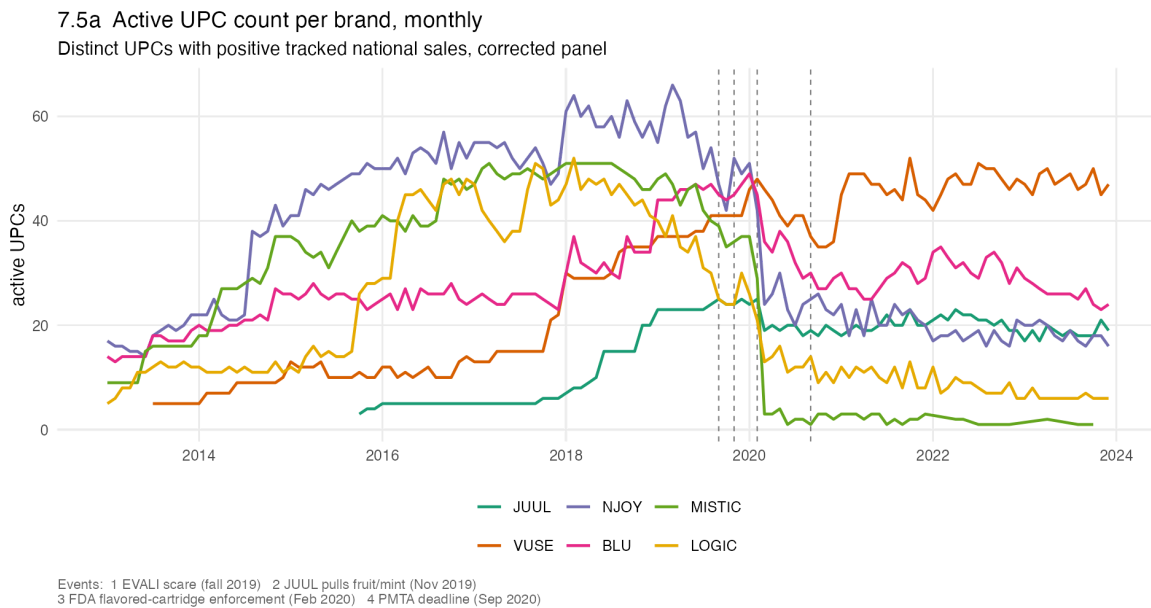
Pooled, there is no signature. With brand and calendar-month fixed effects the elasticity is -0.023 (SE 0.024, not significant); with brand and year effects, -0.024 . Drop the time control and it goes to -0.085 ($p < 0.001$), but that coefficient is just absorbing the well-documented market-wide decline in price per mL (roughly \$8 to \$4.4 over the sample), not own-firm learning. Using cumulative mL instead of units gives -0.038 ($p = 0.14$). So once the common time trend is removed, cumulative own output does not predict a firm's price per mL in the pool.

Firm by firm, it is a cigalike-era phenomenon. Restricting to the 12 brands that ever held at least 3% of national mL in a year and estimating within each brand with year effects: every first-generation / cigalike-era brand has a negative slope — MISTIC -0.34 , LOGIC -0.19 , FINITI -0.17 , BLU -0.15 , Nicotek -0.13 , 21st Century Smoke -0.07 , MarkTen -0.04 , several significant — while the three salt-pod-era leaders are flat: NJOY -0.03 , VUSE -0.02 , JUUL $+0.03$, none significant. The residualized scatter shows the same split visually: BLU, LOGIC, and 21st Century slope down within year; JUUL slopes slightly up, VUSE and NJOY are essentially flat.

Verdict. The "accumulated experience lowers cost" mechanism describes the 2013–2017 cigalike incumbents, whose price per mL fell as they scaled — but that decline is also the period's bulk-juice price war, so it is partly a demand/competition story rather than pure own-cost learning, and the two cannot be separated with price data alone. For the pod-era leaders — the firms the EBE model is actually built to explain — there is no learning-curve signature in price per mL at all. If the endogenous state $a_{i,t}$ is meant to be a cost-reducing experience stock, the data support giving it bite for the cigalike cohort and little to none for JUUL/VUSE/NJOY, whose scaling came with flat or rising price per mL. That argues for either a demand-side interpretation of $a_{i,t}$ (experience shifts willingness-to-pay, not marginal cost) or a cost channel identified from something other than price — margins, or input/PMTA compliance costs — rather than the Benkard price-on-cumulative-output curve.

7.5 Product-offering-set (UPC count) dynamics and the nicotine-weighting check

This section uses a corrected brand $\times$ UPC $\times$ month panel built from the raw RMS files with the same nicotine corrections as `bt_month_t2` (it collapses back to `bt_month_t2` to machine precision — see the appendix). 1,483 e-liquid UPCs; 100–463 active nationally in a given month.



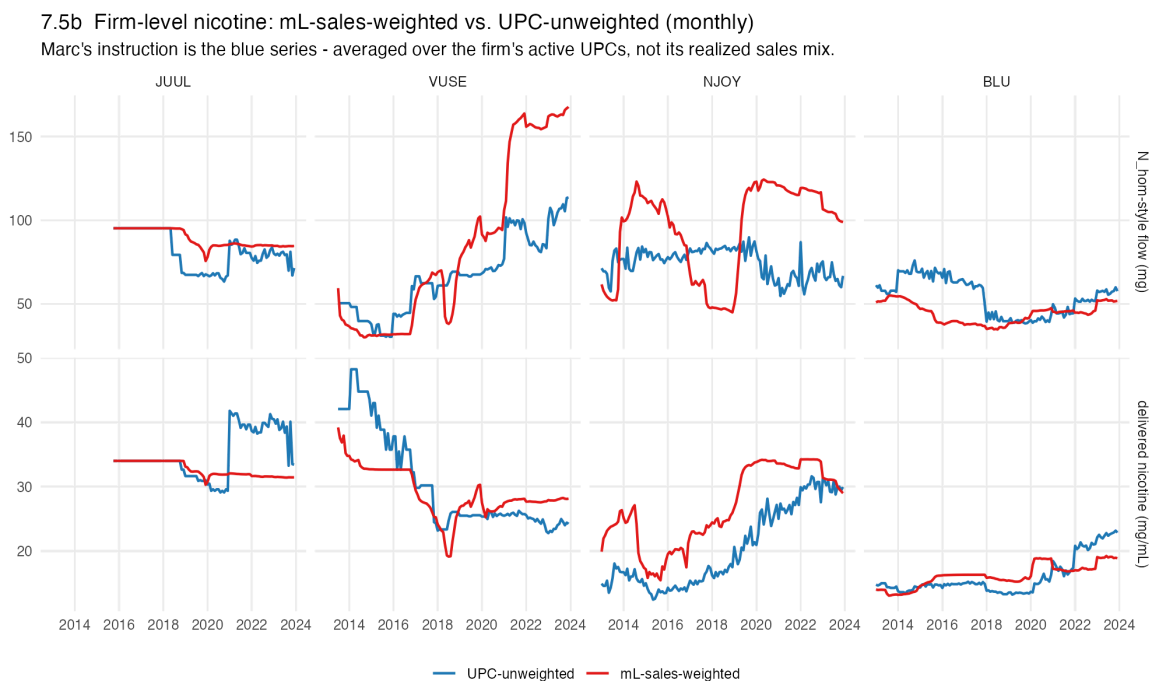
Monthly count of active UPCs (distinct products with positive tracked national sales) per brand. Dashed lines mark the timeline events.

(a) UPC count is not a clean state variable for market position. The offering set is *not* monotone in share. JUUL took the market lead in 2018–2019 (~47% of national mL) while never carrying more than ~25 active UPCs — it won with a deliberately narrow line. NJOY carries the fattest catalog throughout (peaking at 66 UPCs in 2019) and is a persistent #3. The one brand whose UPC count clearly tracks its rise is VUSE, which is also the only incumbent whose catalog *grows* after 2020 (from ~35 to ~52 as it takes the lead). And the single largest movement in every incumbent's catalog is not a firm decision at all: the February 2020 flavored-cartridge enforcement (event 3) is a cliff — MISTIC's active UPC count falls from ~37 to ~2 within two months, LOGIC's from ~30 to ~12, NJOY's from ~50 to ~20, BLU's from ~48 to ~27.

Lead-lag. A distributed-lag regression of the monthly change in national mL share (percentage points) on leads and lags of the change in log UPC count, brand and month fixed effects, SEs clustered by brand: the contemporaneous coefficient is **+0.44 (p = 0.06)** and the one- and two-month leads and lags are all smaller (0.11–0.34) and individually insignificant, roughly symmetric in both directions. Month-to-month, the two change series are essentially uncorrelated pooled ($|\text{corr}| \leq 0.03$ at every lag from -4 to +4); only JUUL shows a weak persistent positive association. So at monthly

frequency UPC-count growth and share growth move *together*, with no robust lead or lag — the monthly first-differences are too noisy to distinguish "expand the catalog, then gain share" from simultaneity. The cleaner pattern is in the levels: catalogs tend to build up through a brand's ascent and rationalize after its peak, consistent with product-line breadth as a slow-moving investment, but the monthly data cannot establish that it *precedes* the demand payoff.

Verdict for 7.5a. UPC count is a poor proxy for the endogenous state $a_{i,t}$: it is non-monotone in market position (JUUL is the decisive counterexample), and its biggest swings are a common regulatory shock (Feb 2020), not firm choices. It is better used as a control or a secondary observable than as the state itself.



Firm-level delivered nicotine, mL-sales-weighted (demand-note style) vs. UPC-unweighted (Marc's instruction), monthly, four largest brands. Top row: N_{hom} -style flow; bottom row: delivered mg/mL.

(b) The weighting choice is not innocuous. Across JUUL, VUSE, NJOY, and BLU, the mL-sales-weighted and UPC-unweighted delivered-nicotine series have a mean absolute gap of **3.9 mg/mL** (correlation 0.84) — roughly 15% of the level, and much more for some brands. The gap is systematic, not noise:

- **NJOY:** mL-weighted delivered nicotine runs at ~33–34 mg/mL in 2019–2023 while the UPC-unweighted average of its active line is ~21–28 — a 6–12 mg/mL gap, because NJOY's sales concentrate in a few high-nicotine pods while its catalog carries many lower-nicotine SKUs. On the N_{hom} -style flow the gap is even wider (mL-weighted ~120 vs UPC-unweighted ~75 in 2020–2022), and the two measures diverge sharply in 2014–2016.

- **VUSE:** the UPC-unweighted series is *higher* early (2013–2014: ~42 vs ~34 mg/mL) and lower later; the mL-weighted N_{hom} flow reaches ~160 mg by 2022 versus ~100 for the UPC-unweighted line.
- **JUUL:** the two track closely while the line is narrow (2015–2018), then diverge — 2021–2023 UPC-unweighted mg/mL rises to ~40 while mL-weighted stays ~32, as JUUL's catalog tilts to 5% SKUs faster than its sales mix does.
- **BLU:** the two series stay within ~1–2 mg/mL through 2020, then the UPC-unweighted line runs ~3–4 mg/mL higher in 2021–2023.

Verdict for 7.5b. Marc's instruction matters. For the two brands the model most needs to get right — VUSE (eventual leader) and NJOY (persistent #3) — the choice between mL-sales weights and UPC-unweighted averaging moves the firm-level nicotine state by 30–50% and can reverse the sign of its change in some periods. The UPC-unweighted series is also the conceptually correct one for a supply-side state: it is a property of the firm's posted product line, not of the current month's realized demand. Recommendation: build $a_{i,t}$ (or its nicotine component) from the UPC-unweighted series, and keep the mL-weighted series only as the object that feeds the demand side.

8. What this changes in the model spec

Pulling the four diagnostics together, against the open choices in §6 of the model note:

- **$\bar{N} = 5$ and the fringe (7.1).** Fine for a pod-era calibration (2019–2023). For a sample that includes 2013–2018, either raise $\bar{N}$ or add an explicit competitive fringe — the sub-top-5 mL residual is 15–20% then and time-varying.
- **Entry process (7.2a).** One-entrant-per-period is close to the data through 2020 and broken in 2021–2022; a sample ending ~2020 keeps the assumption clean, a longer one needs an entry burst (and should treat the 2021 disposable inflow as partly a coverage change, not pure entry).
- **Exit process (7.2b/c).** Enough clean competitive exits to identify a state-dependent exit rule; carve out MarkTen and MISTIC as parent-company (exogenous) exits. Ownership changes at going concerns (JUUL, NJOY) did not cause exit and should not be modeled as exit shocks.
- **The cost channel / $a_{i,t}$ (7.4).** No learning-curve price signature for the pod-era leaders. If $a_{i,t}$ is a cost-reducing stock, it mostly describes the cigalike cohort. Consider a demand-side reading of $a_{i,t}$, or identify the cost channel off something other than price per mL.
- **UPC count as the state (7.5a).** Don't. Non-monotone in share (JUUL: narrow line, market leader), and its largest moves are the Feb 2020 regulatory catalog cull. Use it as a control, not as $a_{i,t}$.
- **UPC-unweighted vs. mL-weighted nicotine (7.5b).** Build the firm-level nicotine state from the UPC-unweighted series, per Marc. The gap to the mL-weighted version averages ~3.9 mg/mL and is 30–50% for VUSE and NJOY — large enough that it is a real modeling choice, not a normalization.

Appendix. Data and construction

- **Panel.** `smoke_firm_dir/input/bt_month_t2_niccorr_from_full.RData` (brand $\times$ type $\times$ month, corrected nicotine, 2013-01 to 2023-12, 132 months) for 7.1/7.2/7.4;
`smoke_firm_dir/input/upc_month_niccorr_from_full.RData` (brand $\times$ UPC $\times$ month, built by `code/build/5_build_upc_month_niccorr.R` with the same row-level corrections, validated by collapsing back to the brand-type panel) for 7.5.
- **Brand universe.** 76 brand-codes; UNKNOWN (1.9% of mL) dropped from all firm-level statistics.
- **Firm counts (7.1).** Shares are mL-based, computed each month over all identified brands.
- **Entry (7.2a).** First month a brand-code has positive national tracked mL; January 2013 is left-censored (all incumbents "appear" then) and dropped.
- **Top-five tenure (7.2b).** Rank by monthly national mL; "months in top-5" is total, not necessarily contiguous; "age at last top-5 month" is months since first appearance.
- **Learning regression (7.4).**
`feols(log(price_per_mL) ~ log(cum_units) | brand + ym)`, cumulative own units through $t - 1$, SEs clustered by brand; firm-by-firm version uses calendar-year FE within brand, restricted to the 12 brands that ever held $\geq 3\%$ of national mL in a year.
- **UPC count / nicotine (7.5).** Active UPC = distinct `upc12` with positive national units that month. UPC-unweighted nicotine = equal-weight mean of each active UPC's own delivered mg/mL (and, for the flow measure, mean mL per unit) across the firm's line. Lead-lag:
`feols(d_share_pp ~ l(d_log_upc, -3:3) | brand + month)`, clustered by brand.
- **Known artifacts flagged in place.** Mid-2019 open-refill clearance; post-2020 modern-disposable undercoverage; the Feb 2020 flavored-cartridge enforcement as a discrete catalog cull; RMS brand-codes persisting at trace volume after a brand is effectively gone.
- **Files.** Code: `code/analysis/10_ebe_diagnostics.R`,
`code/build/5_build_upc_month_niccorr.R`. Figures and CSVs:
`output/prelim_analysis/firm_dynamics_ebe/`.